

Resource Keynote

Addressing (Dis)engagement and Underachievement for Gifted Learners

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Introduction

People often talk about disengagement and underachievement in the same breath, but what we actually need to discuss are two distinct yet, intricately connected concepts – (dis)engagement and (under) achievement. In this article we seek to tease apart disengagement and underachievement and look at the nuances associated with learner engagement and how this contributes to the phenomenon of underachievement. Both concepts have presented as a paradoxical puzzle to gifted students and their teachers, and parents/carers due to long-held stereotypical beliefs around gifted student achievement and talent actualisation, and the potential realities of underachievement.

As a starting point let us establish what we mean by underachievement. Are we all talking about the same entity? Is it academic underachievement as measured by school grades? Does learning outside school count? Sometimes gifted students may not be performing as expected in school, but outside school may have well established areas of 'achievement', such as in their hobbies, entrepreneurship, sporting areas or areas of interest. Take for example, Rob (aged 14 years), with an IQ of 145, who performs averagely at school, mostly Cs for subject-area achievement, and a scattering of Bs and Cs for effort. Outside school he is an accomplished hobbyist with the local lapidary club, cutting and polishing semi-precious stones to design, create and market his own jewellery – a budding entrepreneur! In this case, is Rob still underachieving? We shall leave that to the reader to ponder.

What is underachievement?

There are multiple definitions about what is meant by the term underachievement, but what they all have in common is a discrepancy between measures of a child's ability (or potential) and their actual achievement (Rimm, 2019). The differences in definitions of underachievement arise from how ability and achievement are measured and defined. As we discovered in the example of Rob, some definitions use grades or school-based outcomes to measure achievement, rather than some alternative measures. Other definitions draw on measurements relating to an individual's IQ score in relation to their score on achievement tests; if the IQ score is high (e.g., 145) and the student is achieving averagely at school (e.g., Cs) then there appears to be a discrepancy. As Rimm (2019) points out, these operational definitions can "cause great differences in how and why many gifted students are considered to be underachievers" (p. 3).

Interestingly, when we examine the concept of underachievement through the lens of Gagné's Differentiating Model of Giftedness and Talent (DMGT) (2020a), we are able to apply a broader lens for understanding and possibly intervening in the development of this phenomenon. Gagné (2020a) splits his definition of giftedness and talent into two components – Gifts as *potentialities* (or Aptitudes),

and Talents (or Competencies) as *achievements*.

As such it is worthwhile just briefly providing an overview of Gagné's DMGT (2020a) here. On the left-hand side of Gagné's model there is a list of six *Aptitudes* or *Gifts* under the first heading of *Mental domains* (e.g., *Intellectual*, *Creative*, *Social* and *Perceptual*), and *Physical domains* (e.g., *Muscular* and *Motor Control*). These areas of giftedness occur in the top 10% (minimum threshold) of age-peers. Setting aside for the moment the centre area of Gagné's model (e.g., *Catalysts* and the complex talent developmental process), on the right-hand side we have *Competencies*, or *Talents*, which occur within the top 10% (minimum threshold) of peers (NB: not necessarily age peers). These areas of *Talent* appear as *Competencies* in nine fields of human endeavour – *Academic*; *Technical*; *Science & Technology*; *Arts*; *People Services*; *Management/Sales*; *Business Systems*; *Sports & Athletics*; and *Games*.

Importantly, Gagné talks about the actual talent development process (the 'complex bit in the middle'), as potentially taking a lifetime to actualise as a talent. This is a very important aspect of understanding gifted student underachievement. Do we expect that all gifted students should show their achievement through the narrow curriculum taught and assessed in schools? After all, not all the 'expected' *Fields of Talent* can be directly connected with what is taught and assessed in schools. Perhaps for several of our gifted students, underachievement is part of the complex talent development process, and for these students, achievement (*Talent*), comes well after those 13 years of formal schooling. Nonetheless, to see a gifted child struggling or receiving poor grades at school is discouraging for that child (as well as their teachers and parents). Such catalysts could potentially unknowingly accelerate the progression of underachievement. Before we discuss the complex bit in the middle of Gagné's model (*Catalysts* and the *Developmental Process*), let us turn to briefly explore student engagement and how this might connect to the complexities of underachievement at school.

How might underachievement be tied to engagement for gifted learners?

Interestingly, in 2002 Reis and McCoach had already made some connections to potential causes of underachievement when they noted two main possible causal categories: environmental factors and factors within an individual. In Gagné's complex 'bit in the middle' of his DMGT, his talent development process is very much reliant on two main catalysts – *Environmental* and *Intrapersonal*. This also strongly connects to Reis and McCoach's (2002) theorising about causes of underachievement. Is it feasible that there is a distinct connection between engagement in learning for gifted students and underachievement? Significantly for our discussion, the importance of student engagement in preventing underachievement and school drop-out of gifted students has been well-recognised (Landis & Reschly, 2013).

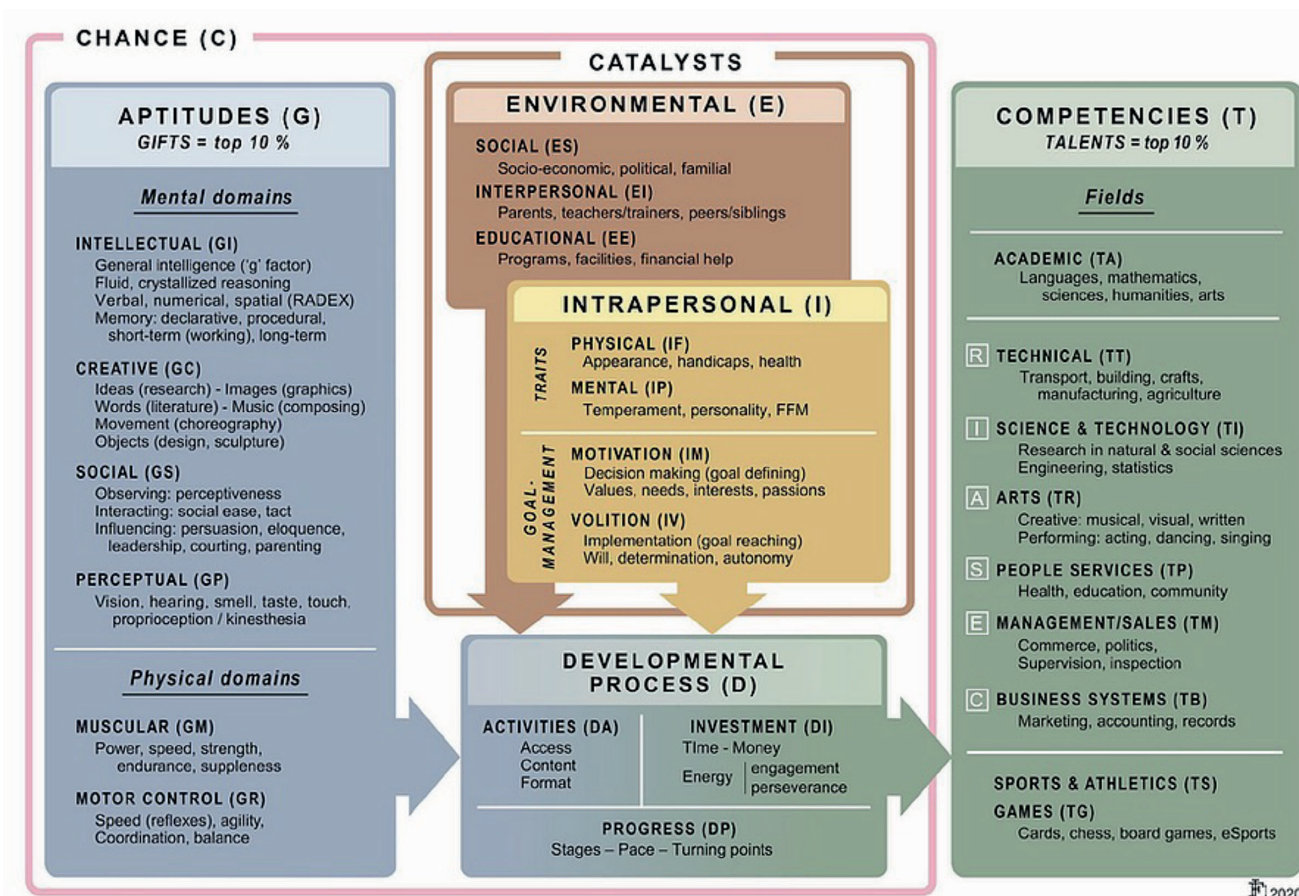


Figure 1 Gagné's DMGT (2020b). Reproduced with permission.

So, what is student engagement?

Student engagement is complex and can be understood by unpacking four different but interconnected dimensions we proposed in the Cyclical Model of Engagement (Ronksley-Pavia & Neumann, 2020). The first dimension is *Behavioural* engagement, relating to a student being present (or not) at school and actively participating (or not), and compliance with school expectations. The second dimension is *Affective* engagement, which relates to elements such as a student's attitude to learning, to school, to peers and teachers, and their interest, enjoyment and identification with the institution of 'school'. The third dimension is *Social* engagement, which connects to a student's sense of belonging in school, connections with peers and teachers and involvement in school-related activities. The fourth dimension, *Cognitive* engagement, relates to a student's learning, such as learning by choice, investment in learning (e.g., through effort and persistence), resilience when faced with educational challenges, a sense of agency in their own learning, and elements relating to self-regulation and metacognition (e.g., setting personal learning goals).

The opposite of engagement, disengagement, is an increasing issue across schools in Australia. The Grattan Institute report (Goss et al., 2017) identified almost 40% of students as being disengaged (Ronksley-Pavia & Neumann, 2020). Estimates of underachievement and disengagement of gifted students range from 10% (Wills & Munro, 2001) to 57% (Peterson & Colangelo, 1996), and the problem is escalating. Gifted students are also becoming increasingly disengaged in the school environment and this is leading to "a loss of potential for both the individual and society as a whole" (Morawska & Sanders, 2009, p. 163).

The complex 'bit in the middle' – addressing disengagement and underachievement

The complex 'bit in the middle' of Gagné's DMGT (2020a) requires that the progression from giftedness (as *potential*) to talent (as *evidenced in achievement*), is facilitated by specific catalysts – *Environmental* and *Intrapersonal*, all of which contribute to the complex Developmental Process. This complex interplay of a multitude of factors through the Developmental Process can enable or disable talent actualisation. It is this process where the development (and impact) of both student engagement and underachievement becomes most evident. These factors speak to the complexities faced when endeavouring to understand and counteract disengagement and underachievement – social (e.g., family), interpersonal (e.g., significant others), educational (opportunities), physical (e.g., disabilities), psychological (e.g., temperament) and engagement elements (e.g., motivation, resiliency, volition learning).

The Developmental Process and the catalysts which impact on this in Gagné's DMGT (Environmental and Intrapersonal) connect quite distinctly with Reis and McCoach's (2002) two categories – environmental factors and individual factors. This adds robustness to the suggestion that any efforts to address underachievement also need to address disengagement at both the environmental *and* individual level.

Learning (and home/community) environments within which teachers (and significant others), make explicit and deliberate efforts to engage gifted learners can begin to address some of the causes of underachievement. In schools, this could be by means of differentiated curriculum and – central to this – personalised learning brought about through understanding the profiles of individual gifted learners.

What are Gifted Learner Profiles?

It is well understood that gifted children are unique individuals. Neihart and Betts' (2010) gifted learner profiles can act as a lens to consider options to help (re)engage underachieving gifted students. The six umbrella profiles of gifted learners suggested by Neihart and Betts (2010) are: Profile I *Successful Learner*; Profile II *Creative Learner*; Profile III *Underground Learner*; Profile IV *At-Risk Learner*; Profile V *Twice-Exceptional Learner*; and, Profile VI *Autonomous Learner* (for a more extensive description see Ronksley-Pavia & Neumann, 2020). These broad gifted learner profiles provide a practical option for educators to better understand how to commence exploring the overarching learning needs of gifted students.

However, to create opportunities to (re)engage gifted students, a much more individual and personalised approach is needed, which can be facilitated (at least in part) by using Neihart and Betts' profile overviews, from which individual profiles can be developed to enable personalised learning for individual students; this may be in the form of an individualised education plan. To begin differentiating instruction for gifted students, educators first need to recognise the individual student's readiness for learning, their interests and understand their individual learner profiles (Tomlinson, 2008). Appropriate opportunities can then be recognised to enable suitable interventions for reengaging individual students, which can then enable more personalised learning. This can begin to address both gifted student engagement and (under) achievement and lead to the development of personalised pedagogical approaches.

What are Personalised Pedagogical approaches?

Engagement is facilitated through a curriculum that is differentiated and personalised to meet student interests and diverse learning needs, and pedagogies that are personalised where the focus is on the individual student. When gifted students are actively engaged in their learning through the four interconnected engagement dimensions, they learn more effectively (Goss & Sonnemann, 2017) and are less likely to disengage from learning (Landis & Reschly, 2013). The challenges associated with personalised pedagogies are well documented (Landis & Reschly, 2013; Ronksley-Pavia, 2019). Central to this challenge is the diversity of the gifted student population. Failure to provide differentiated and personalised instruction can lead to underachievement and disengagement for gifted students.

Brief overview of the key components of re-engagement

Personalised pedagogies for gifted learners facilitate opportunities for developing individual capacity to foster a sense of agency, real-world connections in education, volition learning (learning by choice), self-regulation and leadership capabilities – all of which connect to both environmental and individual potential causes of disengagement and underachievement. Disengagement and underachievement for gifted students starts early on in school, but does not necessarily become visible early on in school. It is therefore vital that any interventions to address these areas begin in primary school.

Our (Re)Engagement Nexus model (Ronksley-Pavia & Neumann, 2020) provides a new conceptual framework and starting point for researching and addressing gifted student disengagement and underachievement across three key areas – four areas of engagement, learner profiles and personalised pedagogies (for a more detailed overview see Ronksley-Pavia & Neumann, 2020).

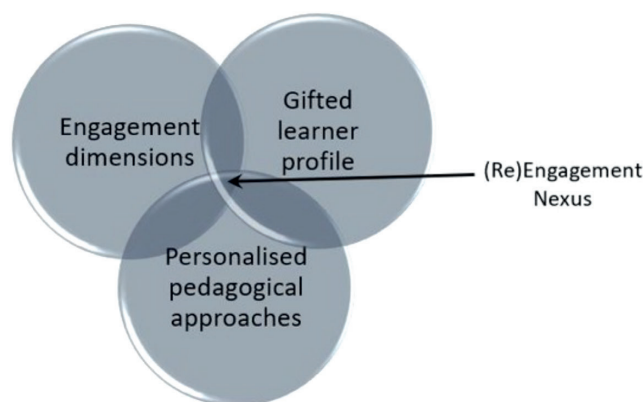


Figure 2 (Re)Engagement Nexus Model (Ronksley-Pavia & Neumann, 2020)

Conclusion

Underachievement is a significant issue for gifted students, which can lead to, or begin with disengagement from learning. The complexities of disengagement and underachievement are compounded by factors which contribute to both – environmental factors (e.g., educational) and intrapersonal issues (e.g., motivation and volition). Together this multiplexity of factors impacts negatively on gifted individuals and society, with many gifted students consequently not having commensurate opportunities to contribute the full extent of their potential to their communities and the economy more broadly. Providing opportunities for personalised learning, through the development of individual gifted learner profiles and suitable pedagogical approaches, can help to ameliorate gifted student underachievement.

To address gifted student (dis)engagement and (under)achievement, it is important to address both environmental and individual (intrapersonal) factors. However, what we first need to agree on is what underachievement actually is (and is not), for gifted learners (i.e., individuals like Rob). It may be semantics, but if we are using Gagné's DMGT to conceptualise and address giftedness, maybe we should be calling the phenomenon *talented student underachievement* – remember giftedness is about potential, talent is about achievement.



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*Simulated Textured Sunflowers after Vincent Van Gogh
- by Alexandria, Year 2,
Sacre Coeur, Glen Iris, Joingy Campus.*